

Heath-Jarrow-Morton model

We will now attempt to model the forward rate $f(t, T)$ instead of the short rate $r(t)$.
Intuition: Short rate is bad for extrapolating for maturities that are far in the future. Say an investor offers a loan that will be executed at $T=10$ years in the future. The investor should likely ask for a higher rate, than what the short-rate predicts.

Continuously compounded forward rate $R(t, T, T+\delta)$:

$$P(t, T) e^{\delta R(t, T, T+\delta)} = P(t, T+\delta) \rightarrow R(t, T, T+\delta) = -\frac{(\log(P(t, T+\delta)) - \log(P(t, T)))}{\delta}$$

Instantaneous forward rate $f(t, T)$:

$$f(t, T) = \lim_{\delta \rightarrow 0} (R(t, T, T+\delta)) = -\frac{\partial \log(P(t, T))}{\partial T}$$

Short rate is just $r(t) = f(t, t)$. Dynamics for $f(t, T)$:

$$f(t, T) = f(0, T) + \int_0^t \alpha(s, T) ds + \int_0^t \sigma(s, T) dW(s)$$

3 HJM conditions $\rightarrow$ non-explosive and progressive processes for drift and volatility to ensure the Bank account $B(t) = e^{\int_0^t r(s) ds}$ remains well-behaved.

Since $\{f(0, t)\}_{t \geq 0}$ is one-to-one with $\{P(0, t)\}_{t \geq 0}$ we can fit the HJM-model to any possible initial term structure. In the short-rate r(t)-models, we could tune drift and volatility to every single maturity, like in HJM.

Example: Suppose we want a special volatility function for the time $T \in [T_a - \epsilon, T_a]$. T_a could be a Fed announcement. Then we could set $\sigma(t, T) = [\sigma_1(t, T), \sigma_2(t, T)]$ with:

$$\sigma_1(t, T) = \sigma_1 e^{T-t}$$

$$\sigma_2(t, T) = \sigma_2 \mathbb{1}_{\{T_a - \epsilon, T_a\}}$$

This is a big advantage - a short rate or ATSM model has a parameter set (H, K, L, θ, ρ) but they are all global parameters, not specifically attached to a maturity.

HJM is infinite-dimensional, meaning that there exists no deterministic function g and Markov process so we can write: $f(t, T) = g(t, T, X(t))$. We can however make it finite dimensional by adding some restrictions on $\sigma(s, t)$.

In the HJM, the ZCP(t) satisfies the SDE:

$$P(t, T) = P(0, T) + \int_0^t P(s, T)(r(s) + b(s, T)) ds + \int_0^t P(s, T) \gamma(s, T) dW(s)$$

$$\text{With } b(s, T) = -\int_s^T \alpha(s, u) du + \frac{1}{2} \|\gamma(s, T)\|^2 \text{ and } \gamma(s, T) = -\int_s^T \sigma(s, u) du$$

Discounting:

$$\frac{P(t, T)}{B(t)} = P(0, T) + \int_0^t \frac{P(s, T)}{B(s)} b(s, T) ds + \int_0^t \frac{P(s, T)}{B(s)} \gamma(s, T) dW(s)$$

HJM is powerful, because we can enforce no-arbitrage through a simple drift restriction in dynamics of $f(t, T)$ under a risk neutral measure $\mathbb{Q}$.

$\mathbb{Q}$ is an ELMM iff:

$$b(t, T) = -\gamma(t, T) \gamma(t, T)^T, \quad \forall t \leq T$$

Which makes $F(t, T)$:

$$f(t, T) = f(0, T) + \int_0^t (\sigma(s, T) \int_s^T \sigma(s, u) du) ds + \int_0^t \sigma(s, T) dW^*(s)$$

$$\frac{P(t, T)}{B(t)} = P(0, T) e^{\int_0^t (\gamma(s, T) \cdot W^*(s)) ds}$$

Proof:

$$\int_0^T \alpha(s, u) du = \frac{1}{2} \|\gamma(s, T)\|^2 - b(s, T)$$

$$\alpha(s, T) = \gamma(s, T) \frac{\partial \gamma}{\partial T} - \frac{\partial b}{\partial T}$$

$$\alpha(s, T) = -\gamma(s, T) \sigma(s, T) - \sigma(s, T) \gamma(s, T)^T$$

$$\text{Girsanov: } dW^* = dW - \gamma(t, T) dt$$

Substitute into forward rate SDE:

$$f(t, T) = f(0, T) + \int_0^t \alpha(s, T) ds + \int_0^t \sigma(s, T) dW(s)$$

$$= f(0, T) + \int_0^t \left[\int_s^T \sigma(s, u) du - \sigma(s, T) \gamma(s, T)^T \right] ds + \int_0^t \sigma(s, T) (dW^*(s) + \gamma(s, T)^T ds)$$

$$= f(0, T) + \int_0^t \sigma(s, T) \left[\int_s^T \sigma(s, u) du \right] ds + \int_0^t \sigma(s, T) dW^*$$

Conclusion: In the HJM-model, the forward rate $f(t, T)$ is fully determined by $\sigma(t, T)$ under the $\mathbb{Q}$ -measure.

Fubini's theorem:

If $\phi(t, s)$ is progressive with $\sup_{t \leq T} \|\phi(t, s)\| < \infty$ then:

$$\int_0^T \left(\int_0^s \phi(t, s) dW(t) \right) ds = \int_0^T \left(\int_0^s \phi(t, s) ds \right) dW(t)$$

Leibniz rule:

$$d \left(\int_0^t f(u, s, t) ds \right) = \left[f(u, t, t) + \int_0^t \frac{\partial}{\partial t} f(s, t) ds \right] dt$$

Leibniz rule for Stochastic integrals:

$$d \left(\int_0^t f(u, s, t) dW(s) \right) = f(u, t, t) dW(t) + \left(\int_0^t \frac{\partial}{\partial t} f(u, s, t) dW(s) \right) dt$$

Ho-Lee model:

$$dr(t) = b(t) dt + \sigma dW^*(t) \Leftrightarrow r(t) = r(0) + \int_0^t b(s) ds + \sigma W_t^*$$

Simply match drift and variance:

$$\sigma(s, t) = \sigma \rightarrow \int_0^t (\sigma(s, t) \int_s^t \sigma(s, u) du) ds = \frac{\sigma^2}{2} t^2$$

$$r(t) = f(t, t)$$

$$r(0) = \int_0^t b(s) ds = f(0, 0) + \frac{\sigma^2}{2} t^2$$

$$b(t) = \frac{\partial f(0, t)}{\partial t} + \sigma^2 t$$

$$r(t) = f(0, t) + \frac{\sigma^2}{2} t^2 + \sigma W_t^*$$

This makes the Bank account:

$$B(t) = e^{\int_0^t r(s) ds} = e^{\int_0^t [r(0) + f(s, s) - f(s, 0) + \frac{\sigma^2}{2} s^2 + \sigma W_s^*] ds}$$

$$= e^{\int_0^t [f(0, s) + \frac{\sigma^2}{2} s^2 + \sigma W_s^*] ds}$$

$$= \frac{1}{P(0, 0)} e^{-\frac{\sigma^2}{2} t^2 + \int_0^t \sigma W_s^* ds}$$

Which makes the ZCP:

$$P(t, T) = \mathbb{E}_t \left[\frac{B(t)}{B(T)} \right] = \frac{P(0, T)}{P(0, t)} e^{\frac{\sigma^2}{2} (T-t)^2} \mathbb{E}_t \left[e^{\int_t^T \sigma W_s^* ds} \right]$$

$$P(t, T) = \mathbb{E}_t \left[\frac{B(t)}{B(T)} \right] = \frac{P(0, T)}{P(0, t)} e^{\frac{\sigma^2}{2} (T-t)^2} \mathbb{E}_t \left[e^{\int_t^T \sigma W_s^* ds} \right] = e^{\mu + \frac{\sigma^2}{2} t^2} \text{ for } X \sim \mathcal{N}(\mu, \sigma^2).$$

How to evaluate expectation? $\mathbb{E}_t[e^X] = e^{\mu + \frac{\sigma^2}{2} t^2}$ for $X \sim \mathcal{N}(\mu, \sigma^2)$.

$$\int_0^T W^* = \left[(T-t) W^*(t) - \int_t^T (T-s) dW_s^* \right] = (T-t) W^*(t) - \int_t^T (T-s) dW_s^*$$

$$\frac{1}{\sigma} \mu = \mathbb{E} \left[\int_t^T (T-s) dW_s^* \right] = \int_t^T (T-s) dW_s^* = (T-t) W^*(t)$$

$$\frac{1}{\sigma} \sigma^2 t = \mathbb{E} \left[\int_t^T (T-s) dW_s^* \right] = \mathbb{E} \left[\int_t^T (T-s) ds \right] = \frac{1}{2} (T-t)^2$$

So we get:

$$P(t, T) = \frac{P(0, T)}{P(0, t)} e^{\frac{\sigma^2}{2} (T-t)^2} + \sigma (T-t) W^*(t) + \frac{\sigma^2}{2} t^2$$

$$= \frac{P(0, T)}{P(0, t)} e^{r(t)(T-t) + f(0, t)(T-t) - \frac{\sigma^2}{2} t(T-t)^2}$$

Only works for Gaussian models like Vasicek or Ho-Lee

Proof: HJM is general
Con: HJM is very computationally inefficient. We have to recalculate $\int_0^t \sigma(s, T) ds$ for each t .

Fix: Enforce Markovianity $\sigma(s, t) = \eta(s) e^{\int_s^t \kappa(s, u) du}$

$\eta(t)$ is a progressive process (it can depend on state variables).

$\kappa_t(u)$ is a deterministic, real-valued function that controls mean-reversion.

Now let's define two different variables:

$$y_i(t) = \int_0^t \sigma_i^2(s, t) ds \quad (\text{Variance accumulator})$$

$$x_i(t) = \int_0^t \sigma_i^*(s, t) ds + \int_0^t \sigma_i(s, t) dW^*(s) \quad (\text{Stochastic driver})$$

$$\text{with } \sigma^*(s, t) = \sigma(t, T) \int_s^T \sigma(s, u) du$$

Then we get:

$$r(t) = f(0, t) + \sum_{i=1}^d x_i(t)$$

$$f(t, T) = f(0, T) + \sum_{i=1}^d \int_t^T \kappa_i(s, t) ds \quad (x_i(t) + \gamma_i(t, T) y_i(t))$$

$$\text{with } \gamma_i(t, T) = \int_t^T \exp(-\int_t^u \kappa_i(s, s) ds) du$$

Note that:

$$dy_i(t) = (\eta_i(t)^2 - 2\kappa_i(t) y_i(t)) dt, \quad y_i(0) = 0$$

$$dx_i(t) = (y_i(t) - \kappa_i(t) x_i(t)) dt + \eta_i(t) dW^*(t), \quad x_i(0) = 0$$

Which makes the typical simulation scheme:

$$y(t_{k+1}) = y(t_k) + (\eta(t_k, x_k, y_k)^2 - 2\kappa(t_k) y(t_k)) \Delta t$$

$$x(t_{k+1}) = x(t_k) + (y(t_k) - \kappa(t_k) x(t_k)) \Delta t + \eta(t_k, x_k, y_k) \sqrt{\Delta t} Z_{k+1}$$

$$P(t_{k+1}, T) = \frac{P(0, T)}{P(0, t_k)} \exp\left(-\frac{1}{2} \gamma(t_{k+1}, T)^2 y(t_{k+1}) - \gamma(t_{k+1}, T) x(t_{k+1})\right)$$

Andreasen paper:

Setting up the general pricing problem in HJM:

$$V(t, x(t), y(t)) = \mathbb{E}_t \left[e^{\int_t^T r(s) ds} \Phi(x(T), y(T)) \right]$$

Just use Feynman-Kac:

$$\partial_t V - [g(0, t) + \kappa] V + [\kappa(t) x + y] \partial_x V + \frac{1}{2} \eta(t, x, y)^2 \partial_y^2 V + (\eta(t, x, y)^2 - 2\kappa(t) y) \partial_y V = 0$$

$$V(T, x, y) = \Phi(x, y)$$

Solving this PDE is much faster than simulating entire paths.

Caplet example:

$$C_P(0) = \delta E_0 \left[e^{\int_0^T r(s) ds} (F(T, T+\delta) - K)^+ \right]$$

$$= E_0 \left[e^{\int_0^T r(s) ds} (1 - P(T, T+\delta) \tilde{K})^+ \right], \quad \tilde{K} = 1 + \delta K$$

Let $\eta(t, x, y) = \sigma F(t, T, T+\delta)^{\alpha}$, $\alpha \in [0, 1]$ and use ZCP-formula:

$$\Phi(t, x, y) = \left[1 - \tilde{K} \frac{P(0, T+\delta)}{P(0, t)} \exp\left(-\frac{1}{2} \gamma(t, T+\delta)^2 y(t) - \gamma(t, T+\delta) x(t)\right) \right]^+$$